

MOHAMED AMIR ABOUZAIID

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PROFESSIONAL SUMMARY

Artificial Intelligence student at Benha University with hands-on experience in Machine Learning, Deep Learning, and Natural Language Processing. Skilled in building end-to-end ML pipelines, hyperparameter optimization, and deploying predictive models. Experienced in text preprocessing, encoding techniques, and feature engineering for NLP tasks. Seeking an entry-level Machine Learning Engineer role to contribute to scalable AI solutions and data-driven decision-making.

TECHNICAL SKILLS

Machine Learning & AI: Supervised & Unsupervised Learning, Classification, Regression, Clustering, Model Evaluation, Cross-Validation, Feature Engineering, Hyperparameter Tuning (Grid Search, Random Search, Bayesian Optimization), Ensemble Methods, Decision Trees, Random Forest, SVM, XGBoost

Deep Learning: Neural Networks, CNN, RNN, LSTM, Transfer Learning, Computer Vision, Image Classification, Object Detection, Model Optimization

Natural Language Processing (NLP): Text Preprocessing, Tokenization, Stemming, Lemmatization, Stop Words Removal, Text Encoding (Bag of Words, TF-IDF, Word2Vec, GloVe), Sentiment Analysis, Named Entity Recognition, Attention Mechanisms, Transformers

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn, Hugging Face Transformers, MLflow

Programming Languages: Python, C++, SQL

Data & MLOps: Data Preprocessing, Data Visualization, ETL Pipelines, Azure AI Fundamentals, Git, GitHub, Jupyter Notebook, Linux

Databases: MySQL, PostgreSQL

PROFESSIONAL EXPERIENCE

Machine Learning Trainee

Digital Egypt Pioneers Initiative (DEPI) – Microsoft Machine Learning Engineer Track | Dec 2025 – Jul 2026

- Completed intensive 6-month training program focused on Machine Learning Engineering using Microsoft Azure technologies
- Learned to translate raw data from various pipelines into scalable data science models and production-ready ML solutions
- Developed algorithms and programs enabling machines to process incoming data and identify complex patterns
- Gained hands-on experience in MLOps tools, MLflow, and Hugging Face for model deployment and lifecycle management
- Studied Azure AI Fundamentals and Azure AI Engineer Associate curriculum for cloud-based AI solutions
- Completed capstone project applying end-to-end ML pipeline from data ingestion to model deployment

Machine Learning Intern

National Telecommunication Institute (NTI) | Sep 2025 – Oct 2025

- Designed and implemented complete ML pipelines including data preprocessing, model training, and performance evaluation
- Built predictive models and optimized performance using hyperparameter tuning techniques to achieve best-fit solutions
- Analyzed datasets and extracted actionable insights to support data-driven decision-making
- Applied core AI concepts: supervised learning, model evaluation metrics, and optimization strategies

EDUCATION

Bachelor of Computer Science and Artificial Intelligence – Major: Artificial Intelligence

Benha University | 2023 – 2027 | General Grade: Very Good

PROJECTS

AI-Based Connect 4 Game System

- Developed a complete Connect 4 game with three modes: Player vs Player, Player vs AI, and AI vs AI
- Implemented AI opponent using Minimax Algorithm with Alpha-Beta Pruning, achieving 60–70% reduction in node exploration
- Designed multi-level AI difficulty system (Easy, Medium, Hard) based on search depth control and optimization
- Built using Python and NumPy with launcher-worker architecture for process isolation and system stability

Meal Master – Restaurant Management System

- Developed a comprehensive desktop application using Java Swing and MySQL
- Implemented core modules: ordering, billing, inventory management, and database operations
- Gained strong experience in database design, SQL queries, and desktop application development

Digital Image Processing System (MATLAB)

- Built comprehensive image processing system implementing RGB to Grayscale/Binary conversions
- Applied point processing techniques: brightness adjustment, negative transformation, contrast enhancement, gamma correction
- Implemented histogram analysis, equalization, and stretching; convolution-based filtering (mean, median, edge detection)
- Simulated various noise types and designed frequency-domain filters

Crowdfunding Platform (Console-Based)

- Built a console-based crowdfunding system using Python and Django
- Implemented user registration, authentication, project creation, editing, deletion, and date-based search
- Designed secure user authentication and data handling mechanisms within a 5-member agile team

CERTIFICATIONS & TRAINING

- Structuring Machine Learning Projects – DeepLearning.AI / Coursera | coursera.org/verify/00OHQUI76DQ8
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization – DeepLearning.AI / Coursera | coursera.org/verify/LENC1E76WHSZ
- Neural Networks and Deep Learning – DeepLearning.AI / Coursera (authorized by Andrew Ng) | Jun 2026 | coursera.org/account/accomplishments/records/ACVRM0Q9YFZ1
- Getting Started with Deep Learning – NVIDIA
- Machine Learning Certificate – National Telecommunication Institute (NTI)
- Microsoft Machine Learning Engineer Track – DEPI (Digital Egypt Pioneers Initiative), 2025–2026

CORE COMPETENCIES

Machine Learning & Optimization: Hyperparameter tuning, model selection, cross-validation, ensemble learning, performance optimization

Deep Learning: Neural network architecture design, CNN/RNN implementation, transfer learning, computer vision

NLP & Text Processing: Text cleaning, tokenization, vectorization (TF-IDF, Word Embeddings), sequence modeling, transformers

Software Engineering: Modular design, version control (Git), agile methodologies, problem-solving, critical thinking

Soft Skills: Teamwork, Communication, Adaptability, Time Management, Leadership, Creativity

LANGUAGES

Arabic: Native | English: Professional Working Proficiency